ROBBINS DATASETS

SUMMARY

3 datasets were released by Robbins with the similar amount of annotations

- 2012: 384345 annotations
- 2014: 384278 annotations
- 2020: 385049 annotations

To understand the changes introduced between 2012 and 2014, I compared craters in both datasets. The most straightforward approach would have been to match craters using their Crater IDs; however, in 2014 the IDs were recreated from scratch using different criteria, resulting in the loss of a direct correspondence with the 2012 dataset. This made the comparison considerably more difficult.
Nevertheless, more than 302,000 craters from the 2012 dataset have corresponding craters in the 2014 dataset with identical latitude, longitude, and diameter values, representing approximately 80% of the craters in each dataset. In other words, only about 20% of the craters underwent any modifications. Of these 20% (approximately 82,000 craters), the vast majority (around 77,000 craters) experienced small changes in longitude and latitude (<0.004°, corresponding to less than one pixel at our 512x512 resolution) and in diameter (less than a 10% difference relative to the corresponding 2012 crater). The remaining 4,000 to 5,000 craters (approximately 1%) underwent more substantial modifications, either with a displacement of more than one pixel in the crater center or with a diameter change greater than 10%.
Beyond this point, establishing reliable correspondences becomes increasingly difficult without visual inspection, since nearby craters of similar sizes may introduce correspondence errors. In summary, approximately 99% of the craters remain essentially unchanged between the two datasets: 80% show no changes at all, while 19% exhibit only minor modifications.

From 2014 to 2020, the crater IDs were largely preserved, with the addition of a leading zero in the final segment of the identifier (e.g., 16-4-07164 → 16-4-007164). The 2020 dataset contains 1,105 craters with no corresponding entry in the 2014 dataset (unique to 2020). There are not exact matches since an automatic adjustment was performed from 2014 to 2020. 96% of craters from 2014 were slightly modified, < 0.004° in the circle center and <10% diameter difference. Roughly 10k craters had changes that were considered moderate and the remaining 6k craters had huge changes in either center position (>0.02º, roughly more than 5 pixels in our resolution) or diameter (>40%).

Visualizing the raw data and Dataset Columns (2012, 2014, 2020)

```
2012
   CRATER_ID  LATITUDE_CIRCLE_IMAGE  LONGITUDE_CIRCLE_IMAGE  LATITUDE_ELLIPSE_IMAGE  ...
0  01-000000                 84.367                 108.746                  84.750  ...
1  01-000001                 72.760                 164.464                  72.784  ...
2  01-000002                 69.244                 -27.240                  69.241  ...
3  01-000003                 70.107                 160.575                  70.151  ...
4  01-000004                 77.996                  95.617                  77.998  ...
[5 rows x 70 columns]
(384345, 70)

2014
    CRATER_ID  LATITUDE_CIRCLE_IMAGE  LONGITUDE_CIRCLE_IMAGE  LATITUDE_CIRCLE_SD_IMAGE  ...
0  01-1-00003                 79.257                -148.095                     0.001  ...
1  01-1-00004                 78.697                -152.840                     0.000  ...
2  01-1-00012                 77.791                -173.416                     0.001  ...
3  01-1-00013                 76.655                -165.191                     0.001  ...
4  01-1-00014                 76.978                -164.378                     0.005  ...
[5 rows x 71 columns]
(384278, 71)

2020
     CRATER_ID  LAT_CIRC_IMG  LON_CIRC_IMG  LAT_ELLI_IMG  LON_ELLI_IMG  DIAM_CIRC_IMG
0  01-1-000003     79.257599    211.909949     79.257584    211.910021      19.526698  ...
1  01-1-000004     78.696512    207.162344     78.696517    207.162105       4.232291  ...
2  01-1-000012     77.791422    186.586023     77.791443    186.585690       8.542042  ...
3  01-1-000013     76.655040    194.807812     76.655020    194.807973       7.967843  ...
4  01-1-000014     76.976285    195.615795     76.976294    195.616254      20.948878  ...
[5 rows x 34 columns]
(385049, 34)

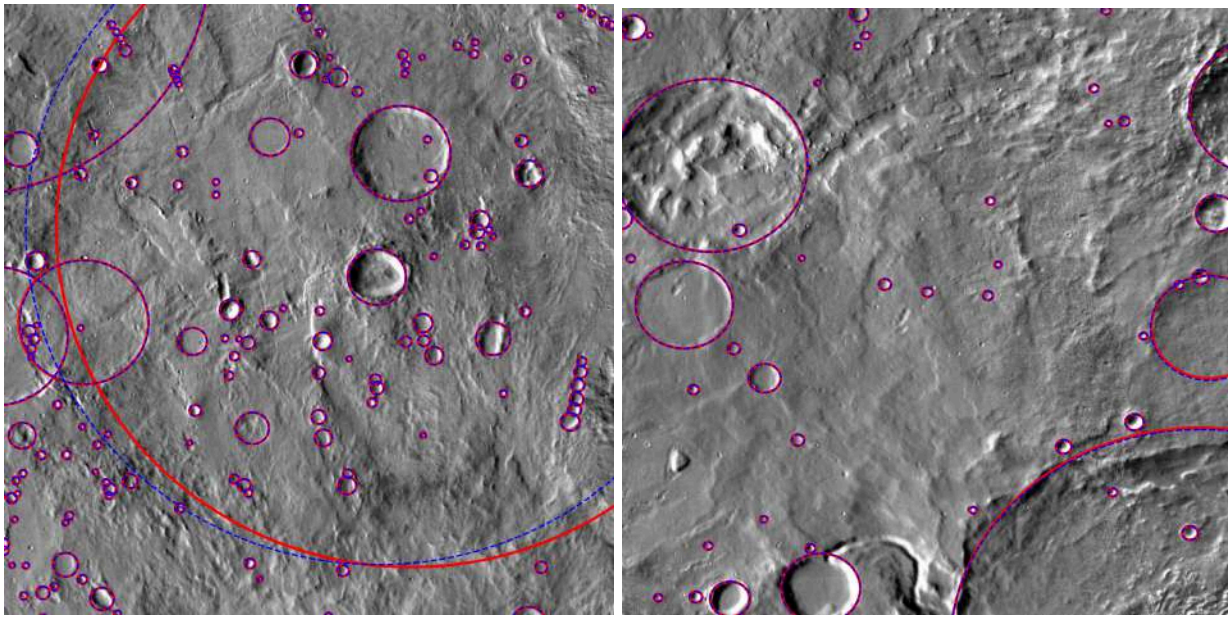
Dataset:  2012                        2014                        2020
Column 1: CRATER_ID                   CRATER_ID                   CRATER_ID
Column 2: LATITUDE_CIRCLE_IMAGE       LATITUDE_CIRCLE_IMAGE       LAT_CIRC_IMG
Column 3: LONGITUDE_CIRCLE_IMAGE      LONGITUDE_CIRCLE_IMAGE      LON_CIRC_IMG
Column 4: LATITUDE_ELLIPSE_IMAGE      LATITUDE_CIRCLE_SD_IMAGE    LAT_ELLI_IMG
Column 5: LONGITUDE_ELLIPSE_IMAGE     LONGITUDE_CIRCLE_SD_IMAGE   LON_ELLI_IMG
Column 6: DIAM_CIRCLE_IMAGE           LATITUDE_ELLIPSE_IMAGE      DIAM_CIRC_IMG
Column 7: DIAM_CIRCLE_SD_IMAGE        LONGITUDE_ELLIPSE_IMAGE     DIAM_CIRC_SD_IMG
Column 8: DIAM_ELLIPSE_MAJOR_IMAGE    DIAM_CIRCLE_IMAGE           DIAM_ELLI_MAJOR_IMG
Column 9: DIAM_ELLIPSE_MINOR_IMAGE    DIAM_CIRCLE_SD_IMAGE        DIAM_ELLI_MINOR_IMG
```

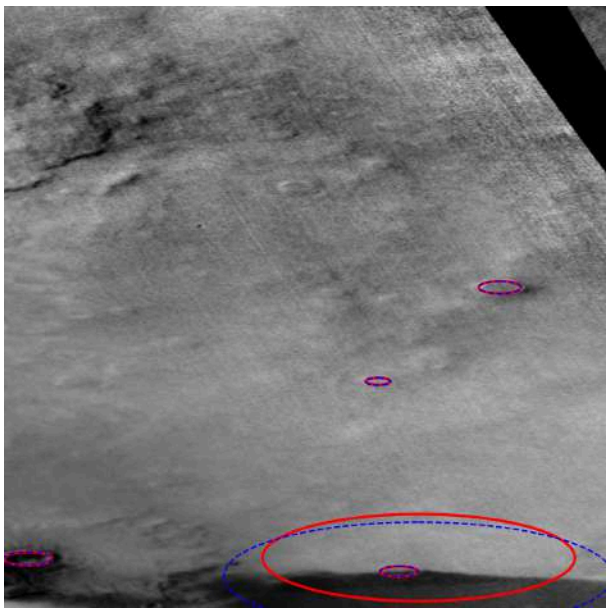
Column 10: DIAM_ELLIPSE_ECCEN_IMAGE	DIAM_ELLIPSE_MAJOR_IMAGE	DIAM_ELLI_ECCEN_IMG
Column 11: DIAM_ELLIPSE_ELLIP_IMAGE	DIAM_ELLIPSE_MINOR_IMAGE	DIAM_ELLI_ELLIP_IMG
Column 12: DIAM_ELLIPSE_ANGLE_IMAGE	DIAM_ELLIPSE_ECCEN_IMAGE	DIAM_ELLI_ANGLE_IMG
Column 13: LATITUDE_CIRCLE_TOPOG	DIAM_ELLIPSE_ELLIP_IMAGE	LAT_ELLI_SD_IMG
Column 14: LONGITUDE_CIRCLE_TOPOG	DIAM_ELLIPSE_ANGLE_IMAGE	LON_ELLI_SD_IMG
Column 15: DIAM_CIRCLE_TOPOG	DIAM_CIRCLE_TOPOG	DIAM_ELLI_MAJOR_SD_IMG
Column 16: DIAM_CIRCLE_SD_TOPOG	DIAM_CIRCLE_SD_TOPOG	DIAM_ELLI_MINOR_SD_IMG
Column 17: DEPTH_RIM_TOPOG	DEPTH_RIM_TOPOG	DIAM_ELLI_ANGLE_SD_IMG
Column 18: DEPTH_RIM_SD_TOPOG	DEPTH_RIM_SD_TOPOG	DIAM_ELLI_ECCEN_SD_IMG
Column 19: DEPTH_SURFACE_TOPOG	DEPTH_SURFACE_TOPOG	DIAM_ELLI_ELLIP_SD_IMG
Column 20: DEPTH_SURFACE_SD_TOPOG	DEPTH_SURFACE_SD_TOPOG	ARC_IMG
Column 21: DEPTH_FLOOR_TOPOG	DEPTH_FLOOR_TOPOG	PTS_RIM_IMG
Column 22: DEPTH_FLOOR_SD_TOPOG	DEPTH_FLOOR_SD_TOPOG	LAY_NUMBER
Column 23: DEPTH_RIMFLOOR_TOPOG	DEPTH_RIMFLOOR_TOPOG	LAY_MORPH1
Column 24: DEPTH_RIMHEIGHT_TOPOG	DEPTH_RIMHEIGHT_TOPOG	LAY_MORPH2
Column 25: DEPTH_SURFFLOOR_TOPOG	DEPTH_SURFFLOOR_TOPOG	LAY_MORPH3
Column 26: PTS_USED_RIM_IMAGE	PTS_USED_RIM_IMAGE	LAY_NOTES
Column 27: PTS_USED_RIM_TOPOG	PTS_USED_RIM_TOPOG	INT_MORPH1
Column 28: PTS_USED_SURFACE	PTS_USED_SURFACE	INT_MORPH2
Column 29: PTS_USED_FLOOR	PTS_USED_FLOOR	INT_MORPH3
Column 30: PTS_USED_LOBE_1	PTS_USED_LAYER_1	CONF
Column 31: PTS_USED_LOBE_2	PTS_USED_LAYER_2	NOTES
Column 32: PTS_USED_LOBE_3	PTS_USED_LAYER_3	DEG_RIM
Column 33: PTS_USED_LOBE_4	PTS_USED_LAYER_4	DEG_EJC
Column 34: PTS_USED_LOBE_5	PTS_USED_LAYER_5	DEG_FLR
Column 35: NUMBER_LOBES	NUMBER_LAYERS	
Column 36: MORPHOLOGY_CRATER_	MORPHOLOGY_CRATER_1	
Column 37: MORPHOLOGY_CRATER_	MORPHOLOGY_CRATER_2	
Column 38: MORPHOLOGY_CRATER3	MORPHOLOGY_CRATER_3	
Column 39: MORPHOLOGY_EJECTA_1	MORPHOLOGY_EJECTA_1	
Column 40: MORPHOLOGY_EJECTA_2	MORPHOLOGY_EJECTA_2	
Column 41: MORPHOLOGY_EJECTA_3	MORPHOLOGY_EJECTA_3	
Column 42: MORPHOLOGY_EJECTA_COMMENTS	MORPHOLOGY_EJECTA_COMMENTS	
Column 43: DEGRADATION_STATE	DEGRADATION_STATE	
Column 44: CONFIDENCE_IMPACT_CRATER	SECONDARY	
Column 45: LAYER_1_PERIMETER	CONFIDENCE_IMPACT_CRATER	
Column 46: LAYER_1_AREA	LAYER_1_PERIMETER	
Column 47: LAYER_1_LOBATENESS	LAYER_1_AREA	
Column 48: LAYER_1_EJECTARAD_EQUIV	LAYER_1_LOBATENESS	
Column 49: LAYER_1_EJECTARAD_REL	LAYER_1_EJECTARAD_EQUIV	
Column 50: LAYER_2_PERIMETER	LAYER_1_EJECTARAD_REL	
Column 51: LAYER_2_AREA	LAYER_2_PERIMETER	
Column 52: LAYER_2_LOBATENESS	LAYER_2_AREA	

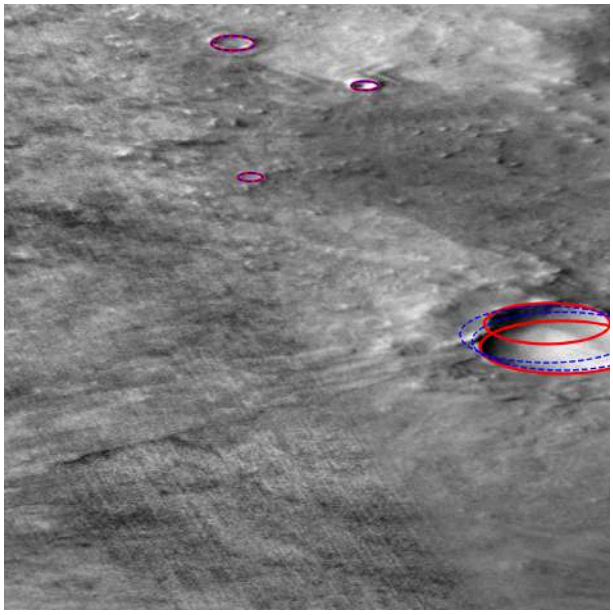
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Column 53: LAYER_2_EJECTARAD_EQUIV      LAYER_2_LOBATENESS
Column 54: LAYER_2_EJECTARAD_REL        LAYER_2_EJECTARAD_EQUIV
Column 55: LAYER_3_PERIMETER            LAYER_2_EJECTARAD_REL
Column 56: LAYER_3_AREA                 LAYER_3_PERIMETER
Column 57: LAYER_3_LOBATENESS           LAYER_3_AREA
Column 58: LAYER_3_EJECTARAD_EQUIV      LAYER_3_LOBATENESS
Column 59: LAYER_3_EJECTARAD_REL        LAYER_3_EJECTARAD_EQUIV
Column 60: LAYER_4_PERIMETER            LAYER_3_EJECTARAD_REL
Column 61: LAYER_4_AREA                 LAYER_4_PERIMETER
Column 62: LAYER_4_LOBATENESS           LAYER_4_AREA
Column 63: LAYER_4_EJECTARAD_EQUIV      LAYER_4_LOBATENESS
Column 64: LAYER_4_EJECTARAD_REL        LAYER_4_EJECTARAD_EQUIV
Column 65: LAYER_5_PERIMETER            LAYER_4_EJECTARAD_REL
Column 66: LAYER_5_AREA                 LAYER_5_PERIMETER
Column 67: LAYER_5_LOBATENESS           LAYER_5_AREA
Column 68: LAYER_5_EJECTARAD_EQUIV      LAYER_5_LOBATENESS
Column 69: LAYER_5_EJECTARAD_REL        LAYER_5_EJECTARAD_EQUIV
Column 70: CRATER_NAME                  LAYER_5_EJECTARAD_REL
Column 71:                              CRATER_NAME
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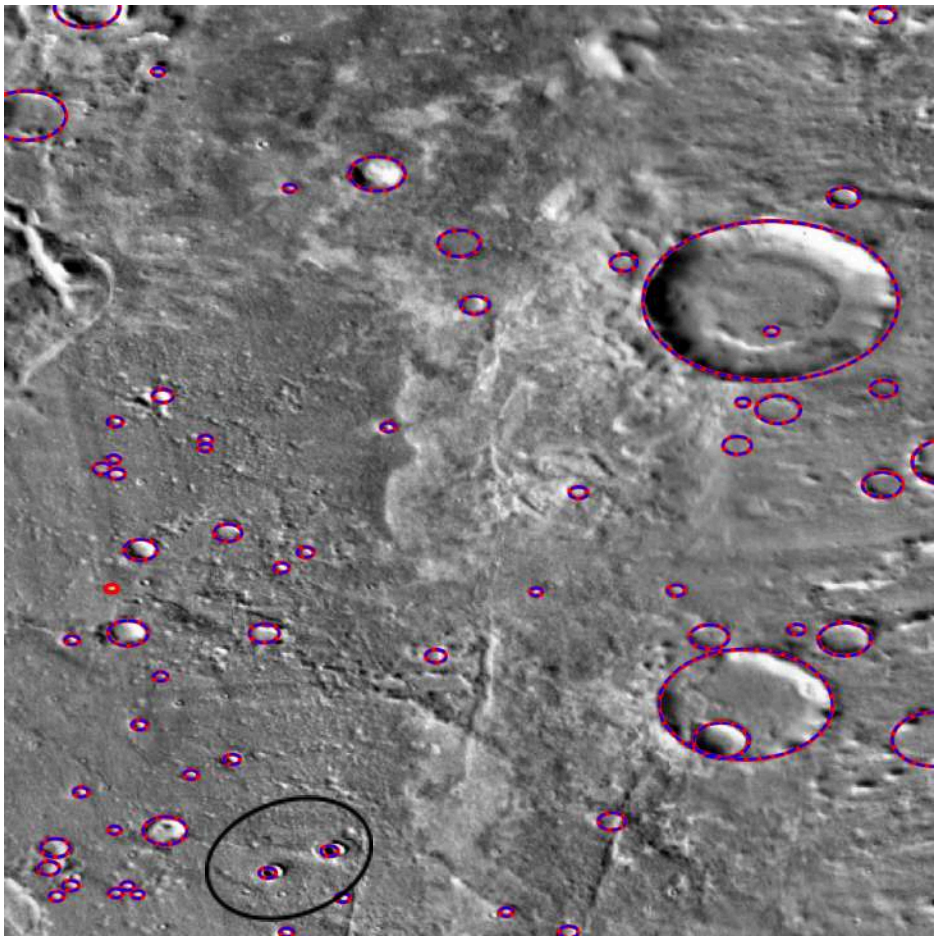
2012 vs 2014

- Red (2012) and Blue (2014). Examples:

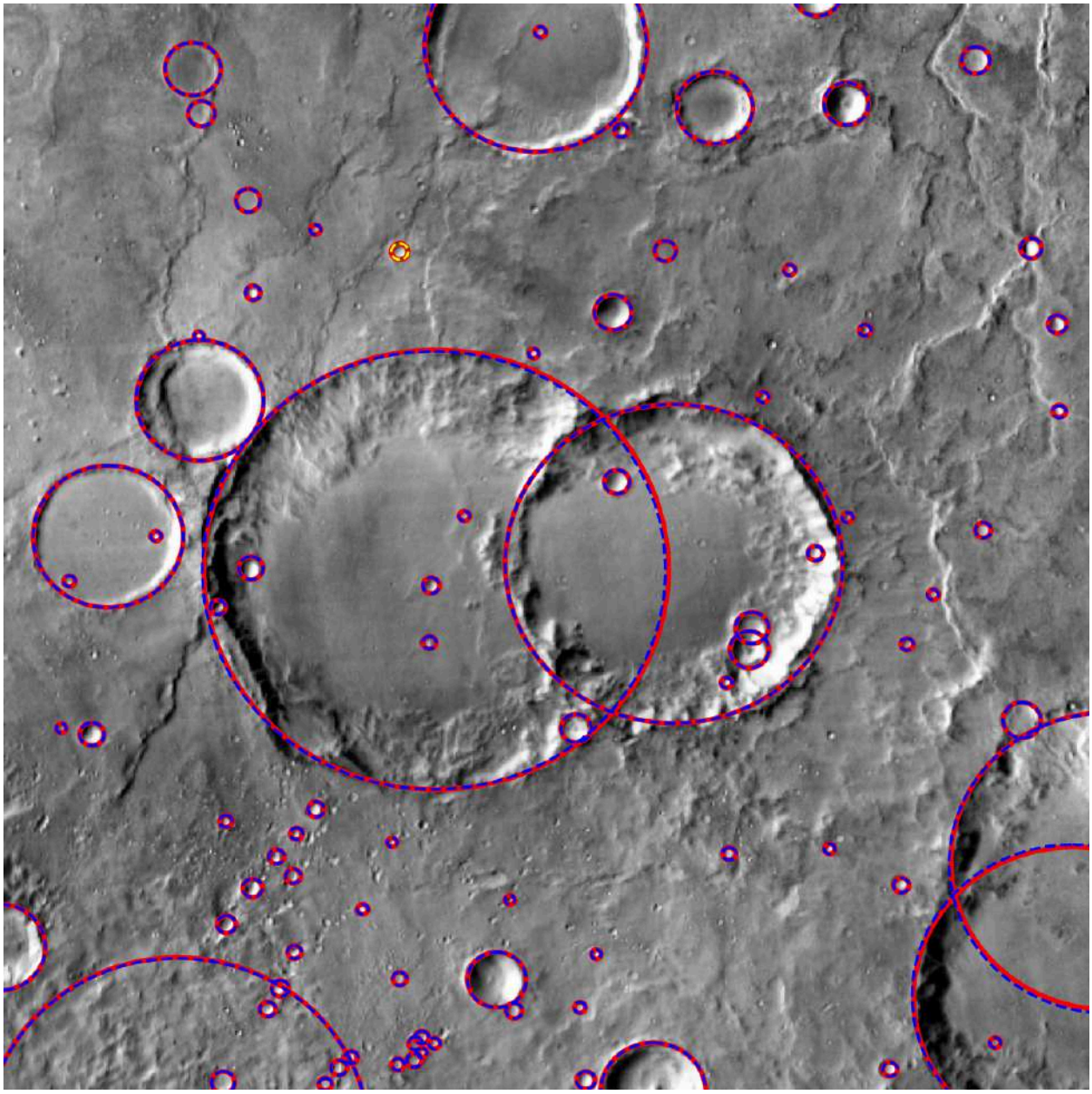








- Even with posterior correction (2014), some annotations are not well aligned with the craters

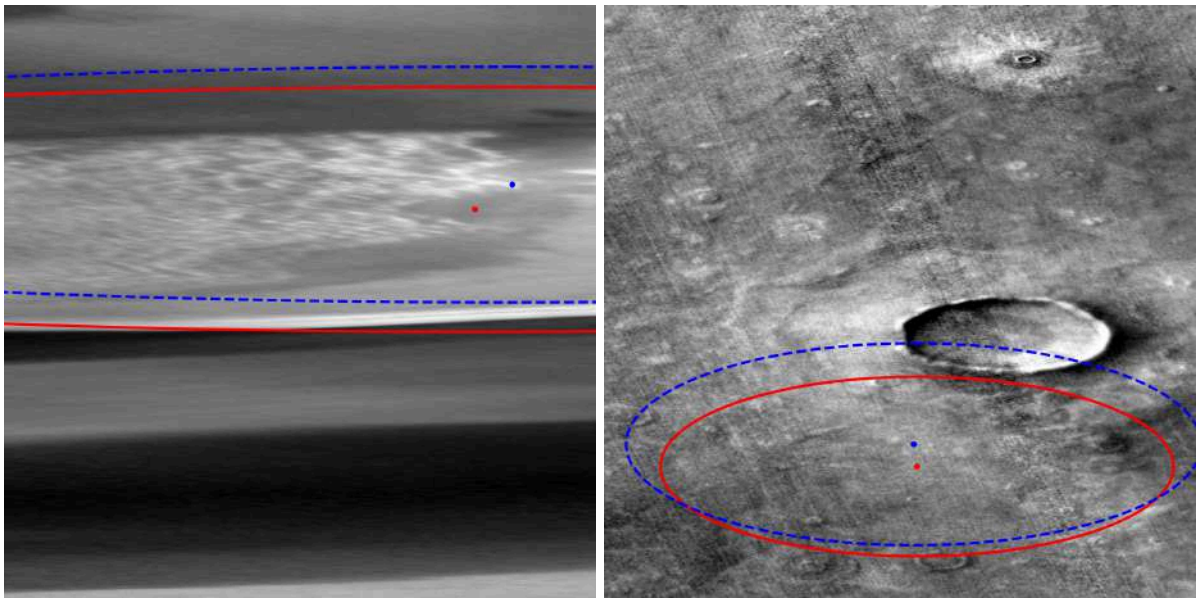


- Moreover, some annotations are duplicated with different IDs on the same dataset (e.g. 2012: Crater ID: 04-003913, Latitude: 32.675, Longitude: -8.033, Diameter: 1.76 and Crater ID: 04-003939, Latitude: 32.675, Longitude: -8.033, Diameter: 1.75)

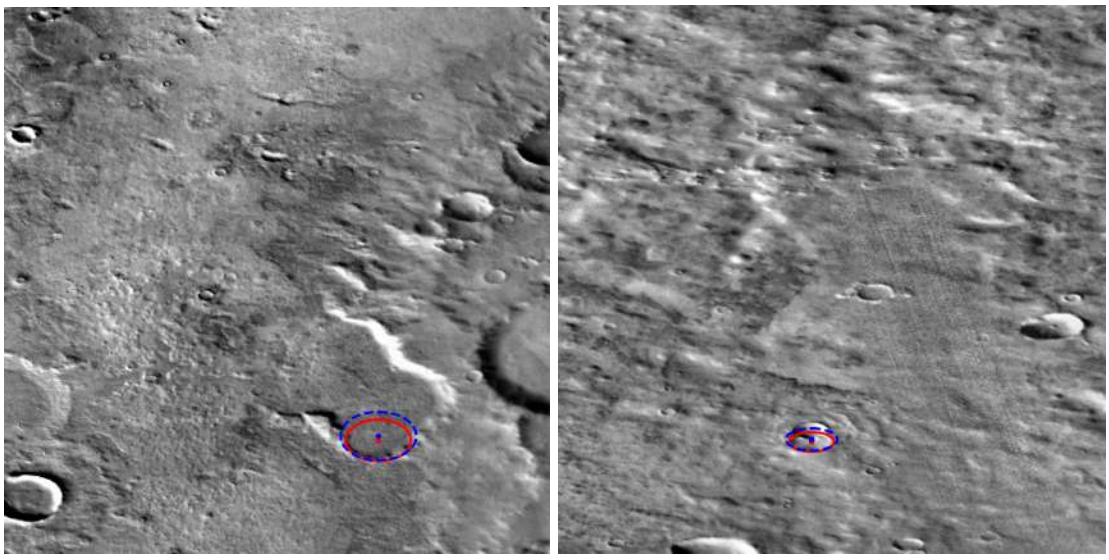
- Since crater-IDs were changed between 2012 and 2014, 1-1 correspondence was lost, which made statistics harder to extract.
- 302K craters (80%) are exactly the same in terms of latitude, longitude and diameter, when comparing between 2012 and 2014 datasets, only 20% were changed
- Of the remaining 20%, the vast majority (19%) has its matching crater very close to itself, in terms both of lat and long of the center of the crater (<0.004º which corresponds 1 pixel max distance in our resolution), and in terms of diameter (less than 10% difference in diameter).

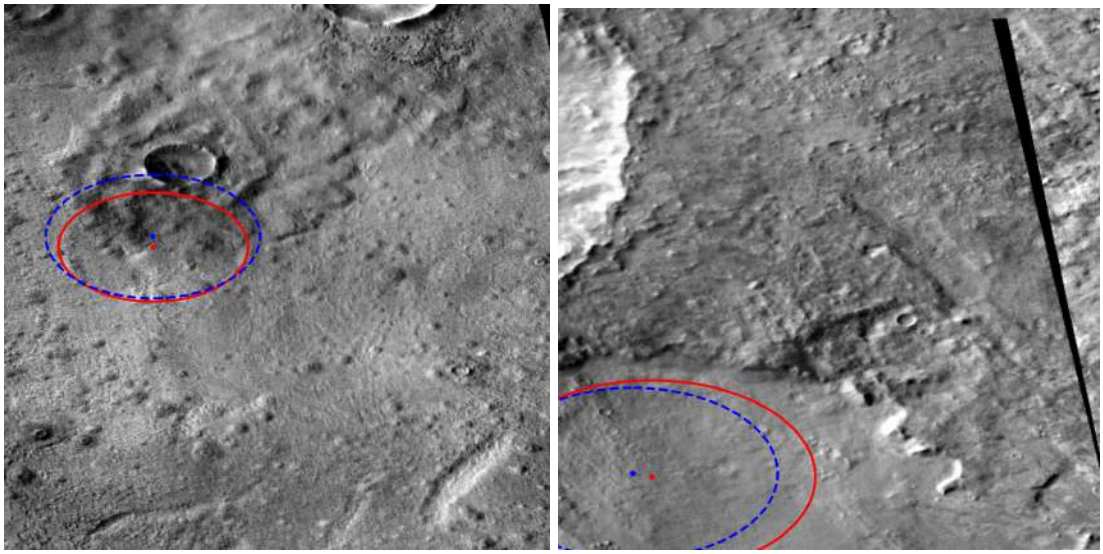
- 1% of craters (4000-5000 craters) are not so easily matched, with greater differences both in center location and in diameter, requiring visual inspection.

Some examples of these 1% craters and their closest annotated crater:



Some annotations seem a bit off, either misaligned or representing an area that is probably not a crater (more common in high latitudes)



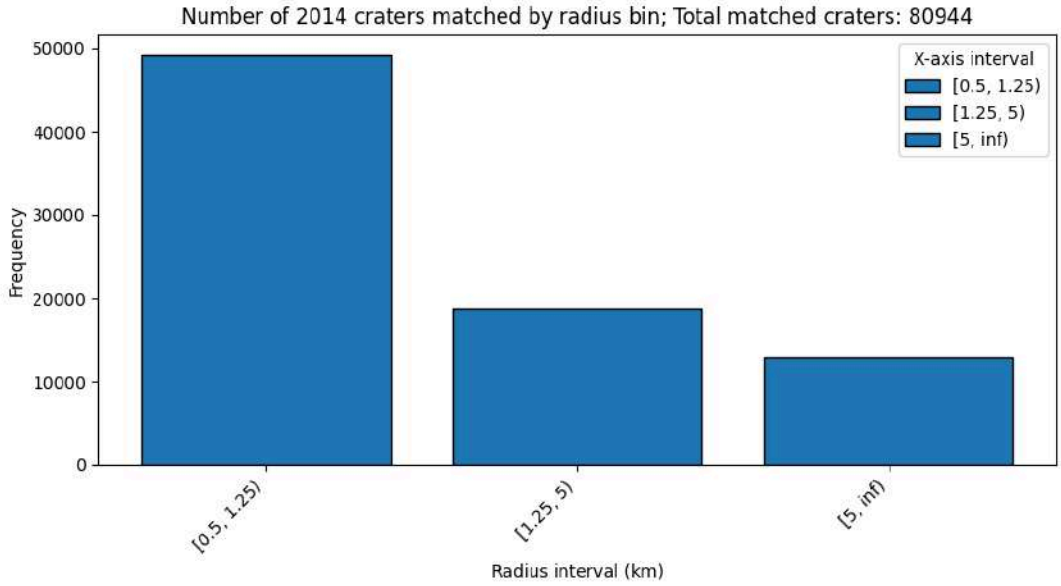


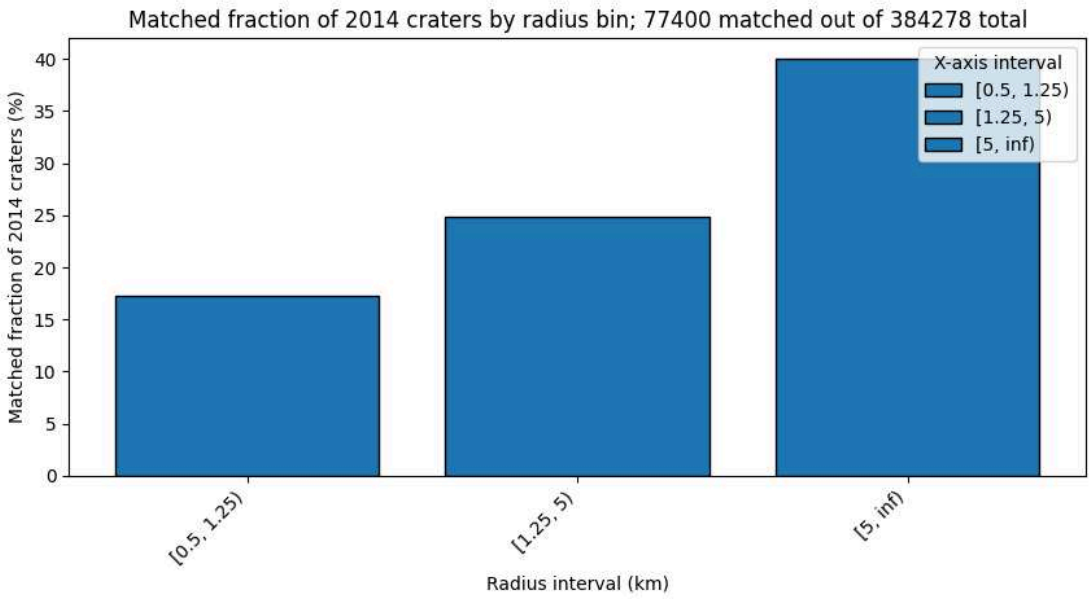
Which craters suffered more updates, smaller or larger craters?

The ones with exact match suffered no update, so they will be discarded for this analysis.

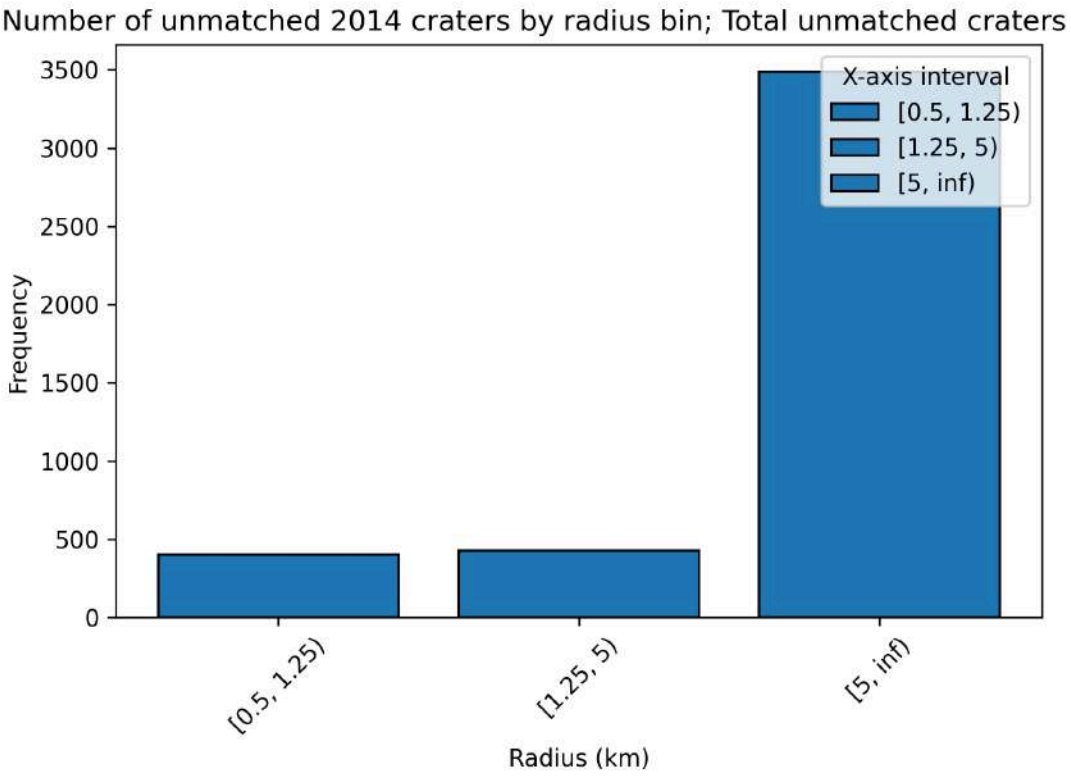
Small modifications (Matched 77k craters).

Starting by the ones with small modifications (77k)
The smallest craters are 0.5 km radius which are around 2 pixels in our resolution (512x512 after downsampling from 1186x1186), therefore the first bins correspond to: **small**, **medium**, **large** (2-5 pixels, 5-20 pixels, 20+ pixels). A higher number of smaller craters was updated however, that corresponds to only 17% of the craters, while 40% of large craters were updated.





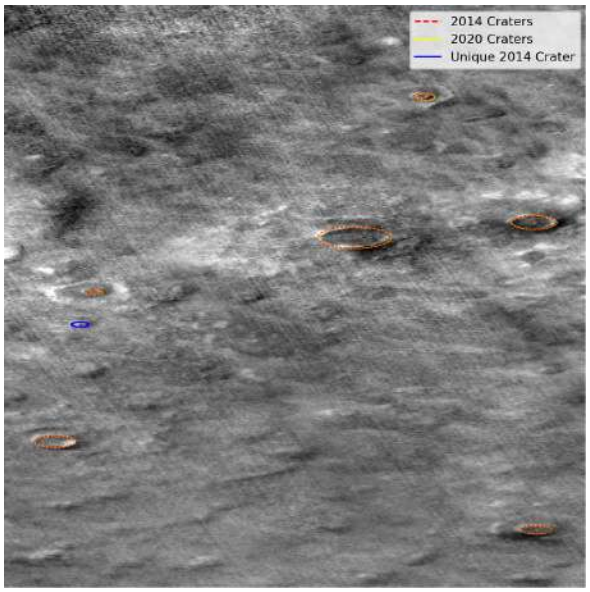
Significant Modifications (Last 4k craters from 2014)



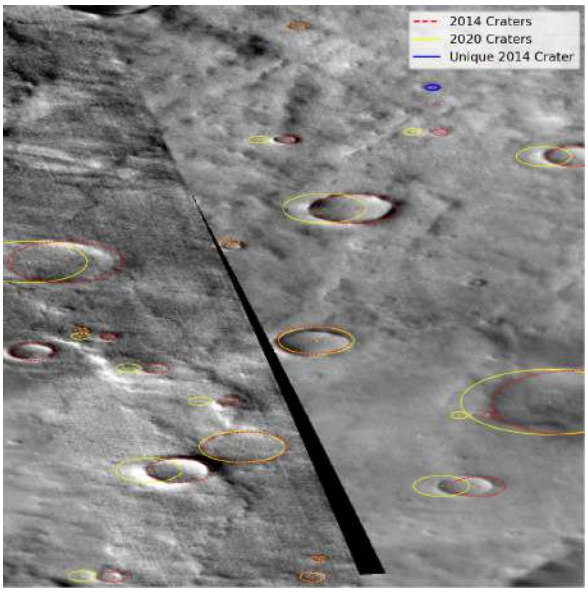
2014 vs 2020

- From 2014 to 2020, ids were kept basically the same, with an extra zero added in the ID number of 2020 (e.g. from 16-4-07164 to 16-4-**0**07164)
- 383944 craters are shared between both datasets (which corresponds to 99+% of the craters)
- The dataset of 2020 has 1105 unique craters (craters with a new/unmatched ID). Conversely, the dataset of 2014 has 334 unique craters.
- Longitudes were converted from -180:180 to 0:360 in the 2020 dataset.

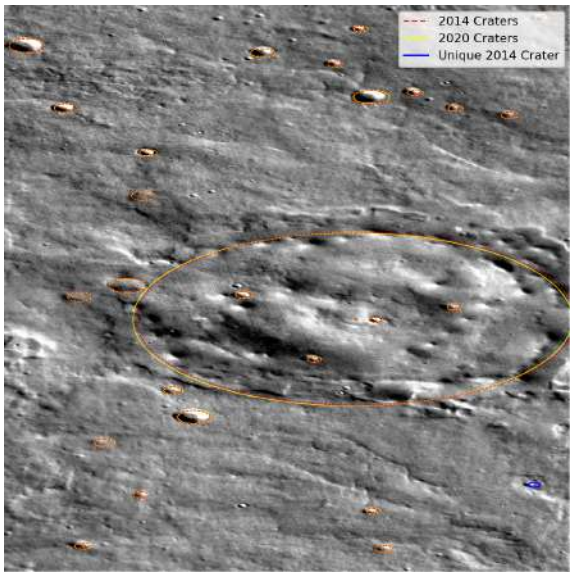
Unique Examples of 2014

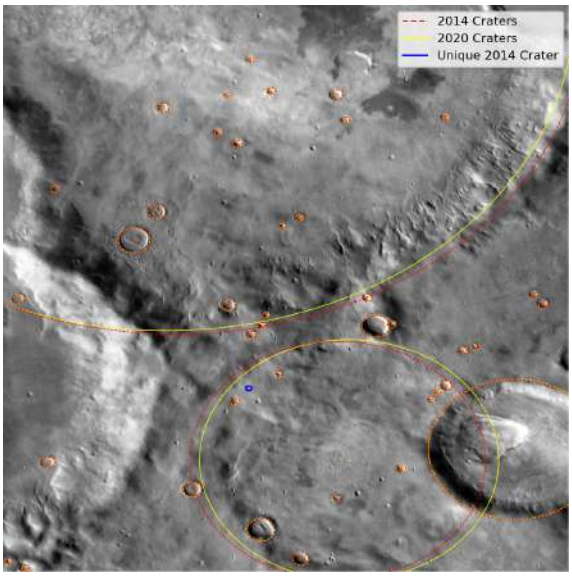


lat_72.0_long_130.0

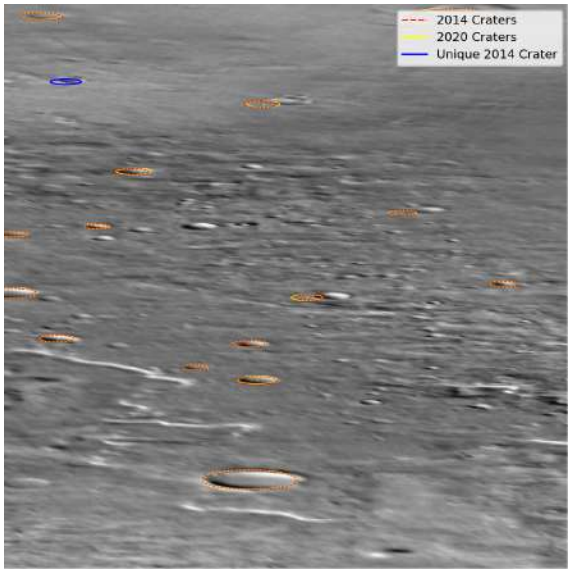


lat_-70.0_long_168.0

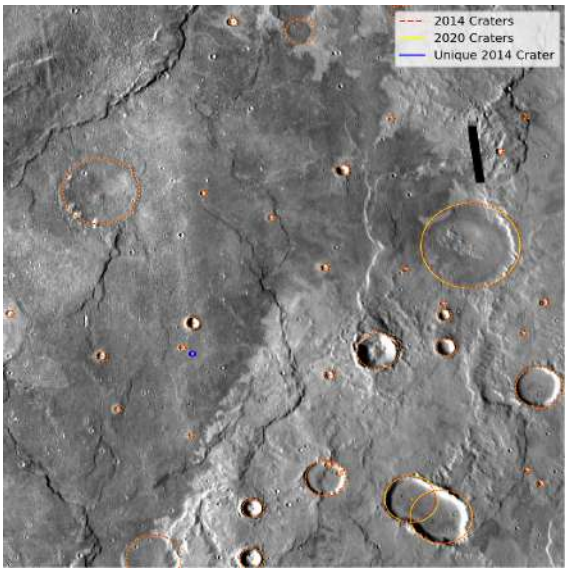




lat_-68.0_long_68.0



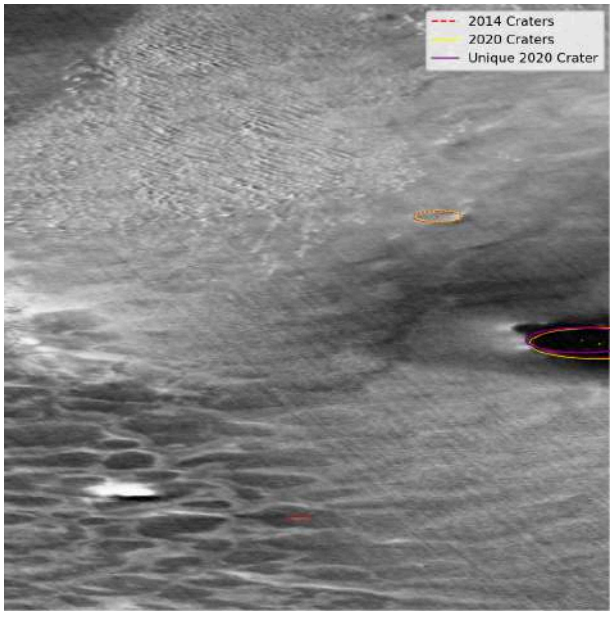
lat_-38.0_long_42.0



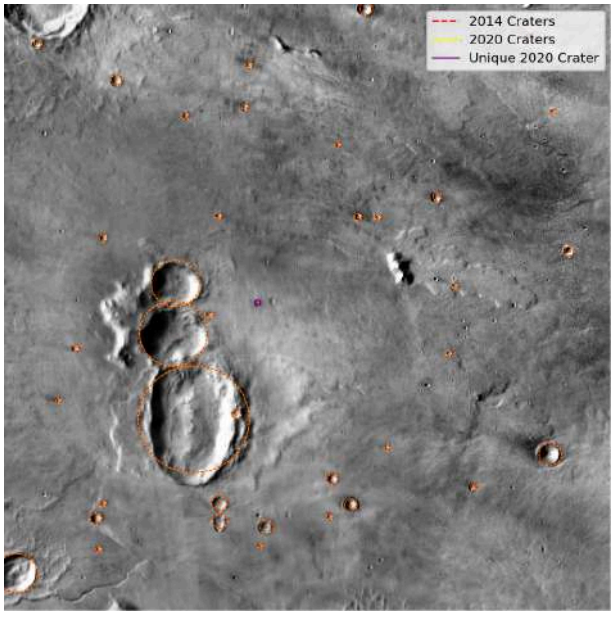
lat_-80.0_long_68.0

lat_-32.0_long_156.0

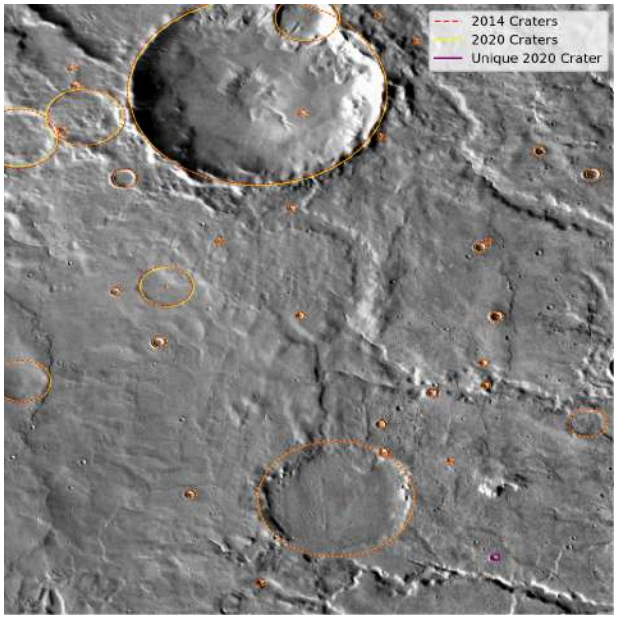
Unique Examples of 2020



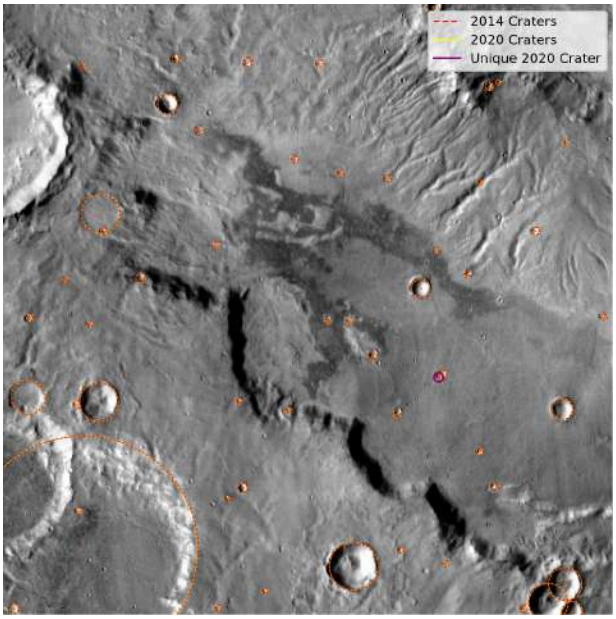
lat_76.0_long_-72.0



lat_14.0_long_76.0.png

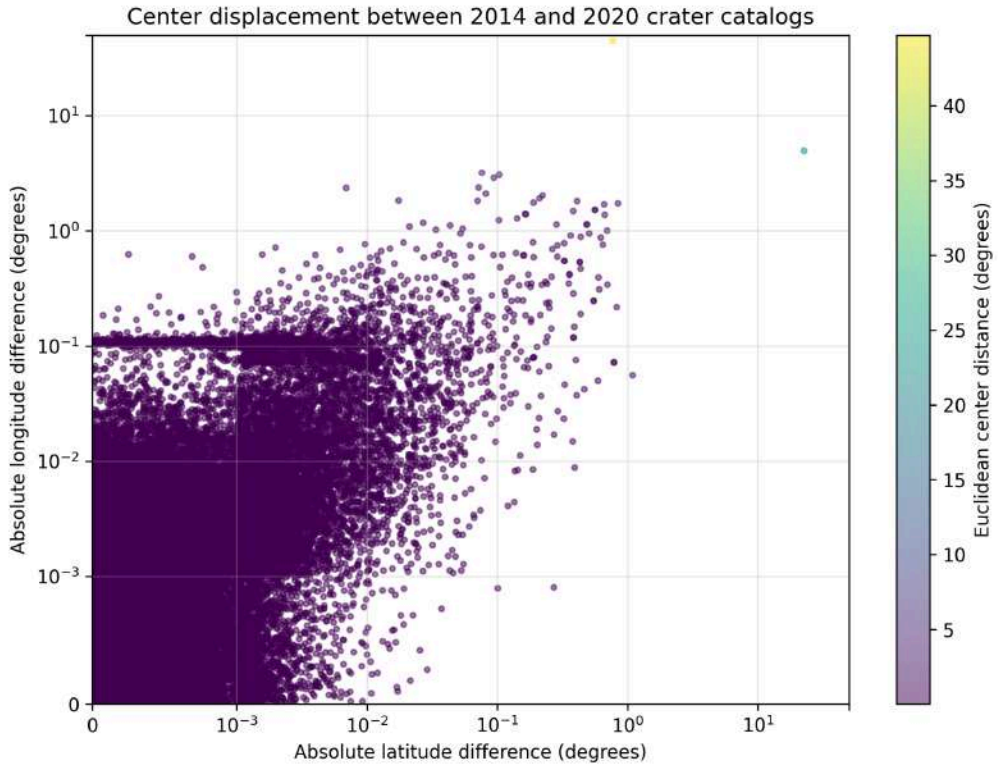


lat_-44.0_long_174.0

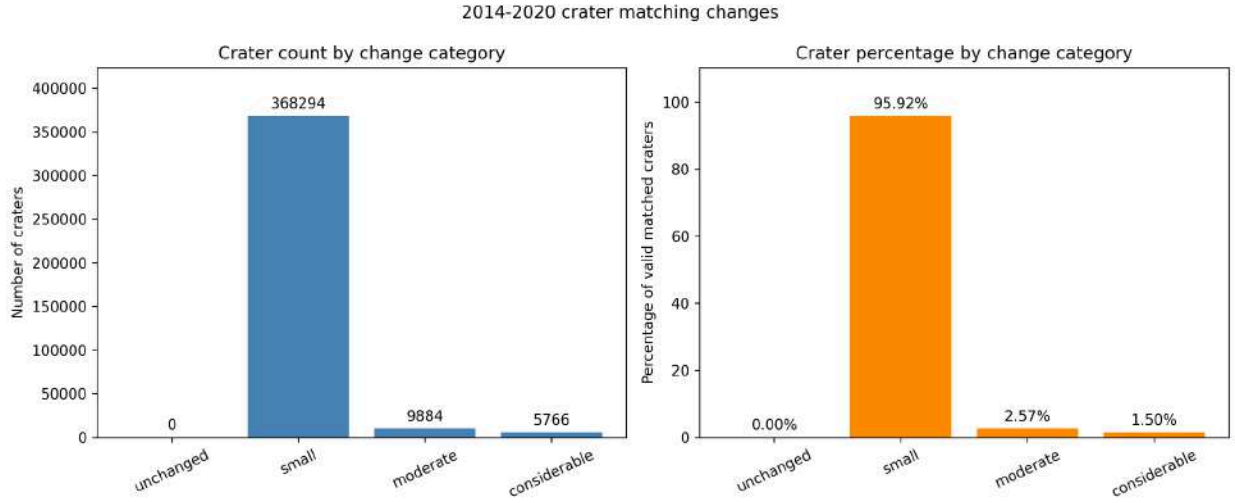


lat_-12.0_long_52.0.png

Center Displacement



Matching differences (Center and Diameter)



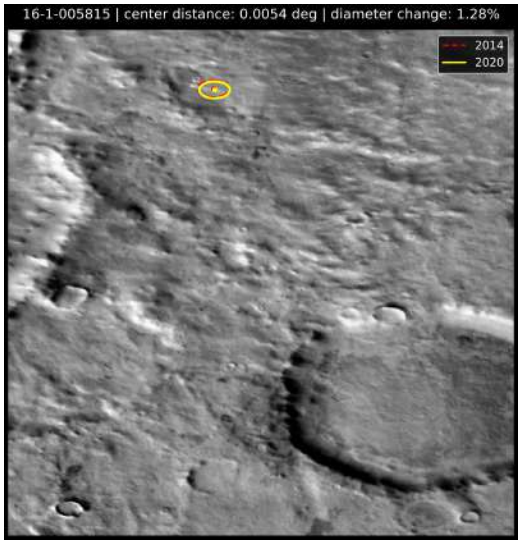
Definitions:

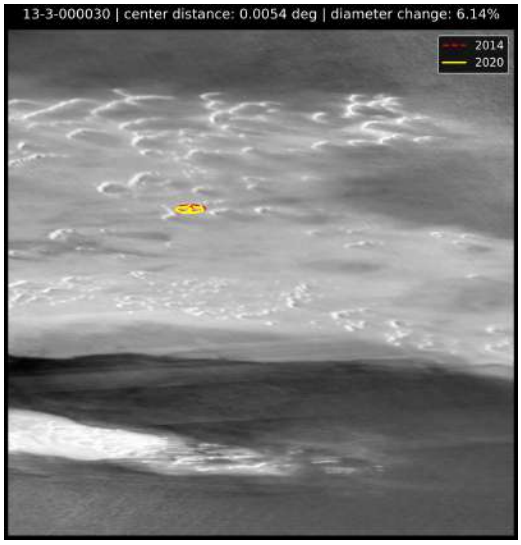
- Unchanged: Latitude, longitude, and diameter are exactly identical in 2014 and 2020.
- Small: Position change is at most 0.004°(~1pixel in our resolution) and diameter change is at most 10%.
- Moderate: Not unchanged, small, or considerable. This means the changes exceed the small thresholds but are not big enough to be *considerable*
- Considerable: Position change is greater than 0.02° (~5 pixels) or diameter change is greater than 40%.

Note: An automatic correction was made from 2014 to 2020, and therefore lat and long changed slightly, even if not manually changed (LAT 79.257 -> 79.2575988 LONG -148.095 -> -148.09005060000004), therefore there are no exact matches.

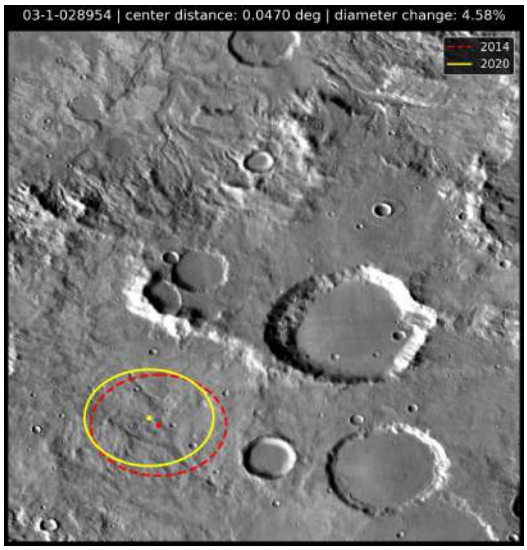
Examples (According to change type)

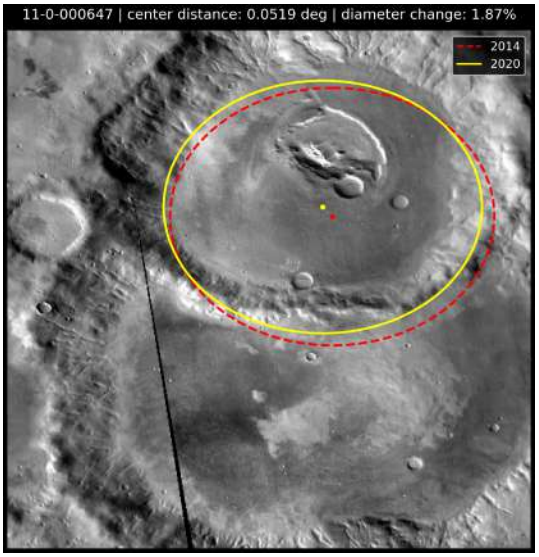
SMALL



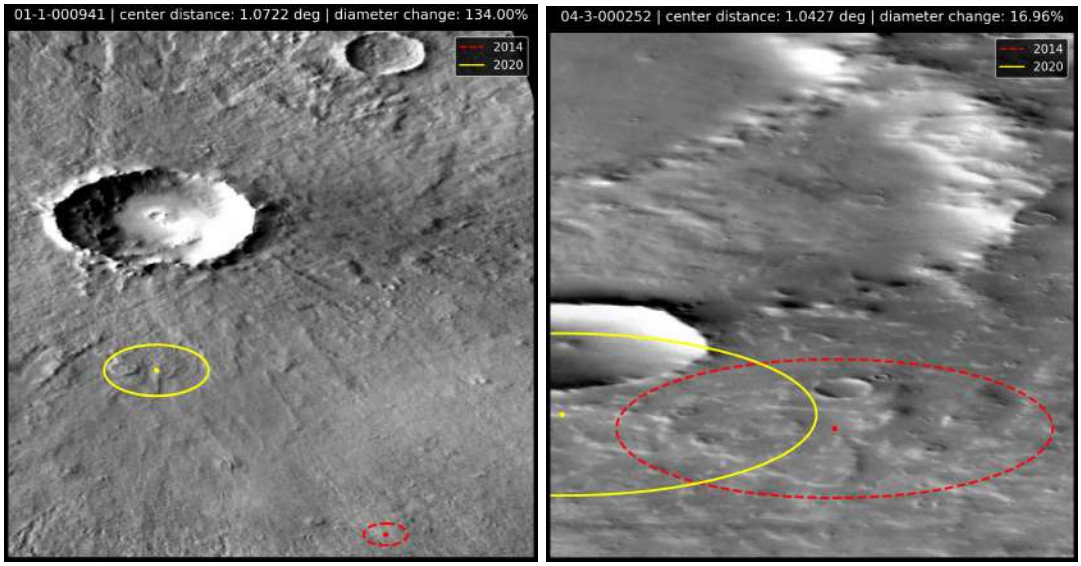


MODERATE

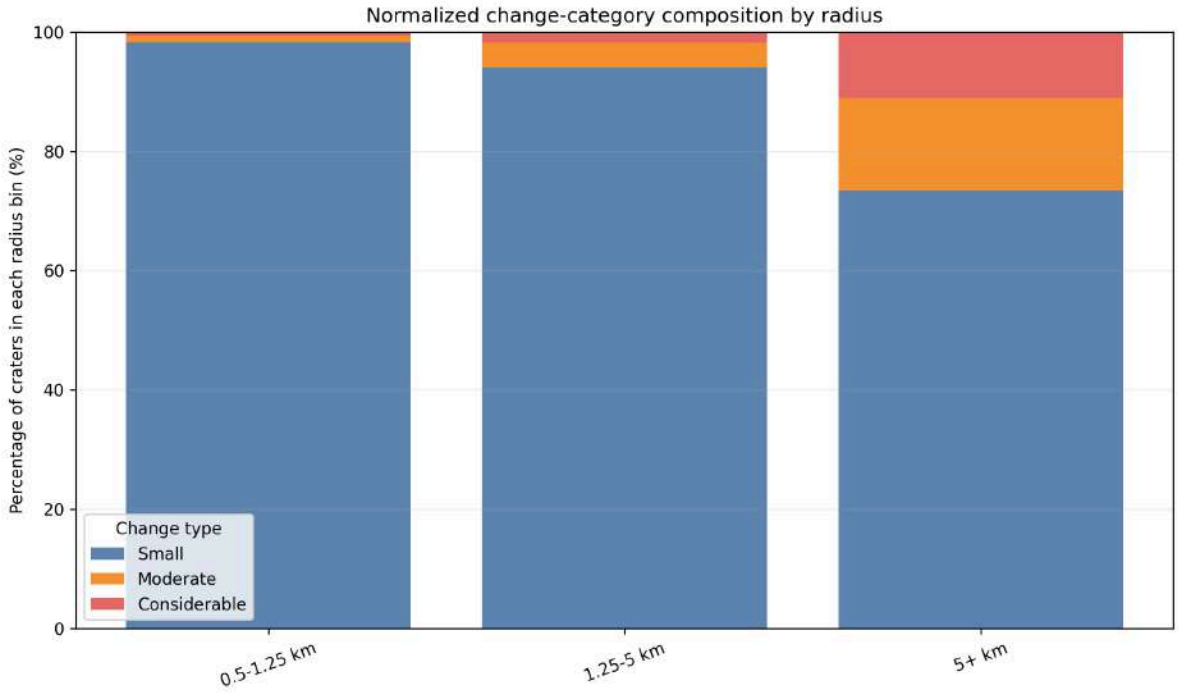




CONSIDERABLE



WHICH CRATERS SUFFERED MORE UPDATES (SMALL, MEDIUM or LARGE)



- This graph was normalized by the amount of craters in each bin, so that it's possible to understand which craters suffered more modifications from 2014 to 2020. The results showed that larger craters were "more" updated i.e. suffered more displacement or diameter increase than medium and small craters.
- Nonetheless, the bias of size cannot be completely dismissed in this analysis since pixel

differences in smaller craters are more significant than large craters, which is the issue when using fixed thresholds for *matching definitions*, and that may have contributed to having larger craters with more significant changes.

CODE (To read the datasets in python)

```
import pandas as pd

df2012 = pd.read_csv(
   "RobbinsCraters_20121016.tsv",
   sep="\t",
   encoding="latin-1",
   low_memory=False,
)

print(df2012.head(20))  # Print the first few rows of the DataFrame to verify
print(df2012.shape)  # Print the shape of the DataFrame to verify

df2014 = pd.read_csv(
   "Combined Catalog - 20140830 (v2).txt",
   sep="\t",
   encoding="latin-1",
   low_memory=False)

print(df2014.head(20))  # Print the first few rows of the DataFrame to verify
print(df2014.shape)  # Print the shape of the DataFrame to verify

df2020 = pd.read_csv(
   "Catalog_Mars_Release_2020_1kmPlus_FullMorphData.csv",
   #df2020 already works with utf-8 default encoding
   low_memory=False
)

print(df2020.head())  # Print the first few rows of the DataFrame to verify
print(df2020.shape)  # Print the shape of the DataFrame to verify
```

MISSING QUESTIONS:

- 2012 vs 2020
- Were there any lats more affected than others? Polar vs Non-Polar
- Should we use circle or ellipse coordinates from the csv? So far we have used circle coordinates.

05/10/2026